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## **The Moral Code: Ethics of Self-Driving Vehicles**

The rise of self-driving cars, also known as autonomous vehicles (AVs), marks a significant milestone in the history of transportation. AVs have the potential to revolutionize transportation. The technology aims to significantly decrease accidents, energy consumption, pollution, and traffic congestion. By utilizing advanced technologies such as sensors, cameras, and complex algorithms, AVs can perceive their surroundings accurately, potentially reducing the number of collisions and saving countless lives (Othman, 2021). However, integrating AVs into society raises ethical concerns, particularly regarding how their algorithms should handle unavoidable accidents.

### **Issue**

The ethical dilemma of AV decision-making is a complex issue with significant ethical implications for the future of transportation. On the roads, autonomous vehicles will encounter situations involving pedestrians, cyclists, and other non-autonomous vehicles, requiring split-second decisions that can impact human lives (Bagloee et al., 2016). The decisions made by AVs will determine who lives and who dies, raising questions about the value of human life and the accountability of AV manufacturers and users. Additionally, these decisions not only affect the outcomes of accidents but also influence public opinion and trust in the technology (Othman, 2021). Ultimately, this issue significantly affects whether autonomous vehicles become widely

adopted and accepted, as it can either build or break public trust in this transformative technology.

## **Background**

Self-driving vehicles have been a concept for several decades, the earliest attempts at autonomous cars date back to the early 1920s (Bagloee, 2016). The ethical dilemma surrounding AV decision-making is not a new concept either. It originates from philosophical experiments like the trolley problem, in which a trolley on a track is headed to kill multiple people. And the only way to save them is to redirect the trolley onto another track, which results in the death of one person. Some argue that the morally correct action is to redirect the trolley, since saving the lives of several individuals is superior to saving one life, others believe that not taking any action would simply be allowing fate to unfold. These complexities have sparked debate and discussion among various stakeholders, each with their own perspectives and concerns.

## **Stakeholder Analysis**

Autonomous vehicles have the potential to transform transportation completely, but before they become widely available on the roads, complex ethical questions must be addressed to ensure that everyone's concerns are heard. There are two key stakeholders in this argument: AV manufacturers and vulnerable road users, such as pedestrians and cyclists.

### **Stakeholder 1: Automobile Manufacturers**

**Values.** Autonomous vehicle manufacturers value technological innovation, market growth, and profitability. They see autonomous vehicles as a revolutionary technology that can transform transportation, but they also recognize the significance of gaining public trust for widespread adoption. To build that trust, they invest heavily in advanced safety technologies like redundant sensors and fail-safes (Padmaja et al., 2023). Ultimately, their financial success hinges on achieving widespread adoption by establishing AVs as the preferred choice for safe and efficient travel.

**Position.** Manufacturers believe that AVs have the potential to improve road safety by reducing human error, increasing transportation efficiency, and creating new revenue streams through the sale and operation of autonomous fleets (Padmaja et al., 2023). These aspects are significant because they directly affect manufacturers' ability to innovate, stay competitive in the market, and meet their business objectives (Othman, 2021). The success of AVs could redefine transportation and reshape the automotive industry.

**Claims.** Manufacturers employ various claims to promote the advantages of AVs. They utilize claims of fact, citing statistics and data on the potential safety benefits of AVs, such as reduced accidents and fatalities due to the elimination of human error (Othman, 2021). They also make claims of cause, arguing that AVs can address challenges like traffic congestion and pollution by optimizing traffic flow and enabling the use of electric vehicles (Bagloee et al., 2016).

Furthermore, manufacturers employ claims of definition, framing AVs as a technological innovation that represents the future of transportation (Padmaja et al., 2023).

## **Stakeholder 2: Road users, pedestrians and Cyclists**

**Values.** Pedestrians and cyclists also place emphasis on a transportation system that effectively reduces the likelihood of accidents and injury. However, they prioritize their safety above all else and value being able to move around their communities without any fear of encountering accidents (Hansson et al., 2021). So, they place importance on infrastructure and regulations that emphasize their safety, such as dedicated lanes and stringent safety standards for autonomous vehicles. According to Bagloee et al. (2016), They also value clear regulations and thorough testing of AVs to ensure their safety when interacting with vulnerable road users.

**Position.** The safety of pedestrians and cyclists will be highly impacted by the implementation of AVs. These vulnerable road users face higher risks in traffic accidents, and the decisions made by AVs in critical situations can have life-or-death consequences for them (Bonnefon et al., 2016). Given the potential life-or-death consequences of AVs' decisions, pedestrians and cyclists advocate for regulations and policies that prioritize their safety. They are particularly concerned about the potential for AVs to prioritize the safety of their passengers over their own.

**Claims.** Pedestrians and cyclists counter manufacturers' claims with their own arguments. They use claims of fact, such as the occurrence of unavoidable incidents involving self-driving vehicles (Hansson et al., 2021) and the potential for AV programming to favor certain lives over others (Bonnefon et al., 2016). Also, they highlight that AVs cannot replicate the communication of human drivers, potentially putting pedestrians and cyclists at risk (Rothenbucher et al., 2016). Human drivers can make eye contact, use hand gestures, or nod their heads to signal their intentions to pedestrians. Additionally, they make claims of cause, stating that if AV algorithms are programmed to prioritize passenger safety, it could directly lead to increased risks for them

(Bonnefon et al., 2016). This prioritization could lead to increased risks for those outside the vehicle.

### **Argument Question**

Should autonomous vehicles be programmed to make life-or-death decisions? AV manufacturers argue in favor, while other road users like pedestrians and cyclists oppose it.

### **Stakeholder Arguments**

Stakeholder 1: Automobile Manufacturers

**Framework.** Driven by a utilitarian framework, AV manufacturers would likely advocate for programming self-driving cars to minimize the overall harm in unavoidable accidents (Bonnefon et al., 2016). Utilitarianism, a consequentialist ethical framework, was introduced in the 18th and 19th centuries by Jeremy Bentham and John Stuart Mill, is based on the principle of maximizing overall happiness or utility, and it proposes that the most ethical choice is the one that produces the greatest good for the majority (Hamlett, 2014). In essence, the utilitarian framework focuses on the outcomes or consequences of actions rather than the intentions behind them (Hamlett, 2014), aiming to achieve the greatest good for the greatest number.

**Rationale.** In the context of AV decision-making, this translates to designing algorithms that prioritize the safety of the majority. For manufacturers, this approach aligns with their goal of maximizing safety. By programming AVs to make such decisions, they contribute to the overall reduction of traffic accidents and fatalities, meeting the expectations for safer transportation (Bagloee et al., 2016). In a study by Faulhaber et al. found that people preferred a utilitarian

approach, choosing to save the greatest number of lives in a simulated accident scenario, even if it meant sacrificing themselves (Othman, 2021). This utilitarian perspective emphasizes the greater good, even if it means sacrificing the few in extreme situations.

**Advocacy.** AV manufacturers advocate for the rapid development and widespread adoption of AVs, potentially with programming that prioritizes minimizing harm. They believe that the potential benefits of reduced accidents and injuries outweigh ethical concerns about individual cases where the AV's decision might seem unfair (Bonnefon et al., 2016). Their goal is to increase the adoption and integration of this technology by maximizing the benefits of AVs.

**Implications.** Manufacturers expect significant gains from widespread AV adoption, driven by increased sales and market share as demand for safer and more efficient transportation grows (Bagloee et al., 2016). Additionally, programming AVs with a utilitarian approach, prioritizing the minimization of overall harm in accidents, could potentially reduce manufacturers' liability (Bonnefon et al., 2016). Which could lead to significant cost savings. However, manufacturers also may face potential losses. Programming AVs with a utilitarian approach could lead to public backlash and negative media attention, especially in cases where passengers are sacrificed to save others (Bonnefon et al., 2016). This could damage their reputation and deter potential buyers. Additionally, they may face challenges with regulations and legal frameworks that may not be fully equipped to handle the complexities of utilitarian decision-making in AVs (Othman, 2021). Such uncertainties could slow down market adoption and reduce predicted profitability.

**Stakeholder 2: Pedestrians and Cyclists.**

**Framework.** From the perspective of pedestrians and cyclists, the ethical framework of care ethics would strongly advocate for prioritizing their safety in unavoidable accident scenarios. Care ethics was developed by feminist philosophers Carol Gilligan and Nel Noddings, and it emphasizes the importance of relationships, empathy, and responsibility in ethical decision-making (Nair, 2014). Care ethics encourages us to put ourselves in the shoes of others, understand their perspectives, and respond with care and concern for their well-being. Care ethics highlights our moral obligation to care for those who are vulnerable or dependent on us (Nair, 2014). This ethical framework urges us to prioritize the safety of pedestrians and cyclists, recognizing their vulnerability on the roads.

**Rationale.** Care ethics would argue that self-driving cars making life-or-death decisions should prioritize empathy and the well-being of the vulnerable, in the case of self-driving vehicles, pedestrians, and cyclists, are vulnerable road users, and should not be subject to AVs' life-or-death decisions. Care ethics emphasizes the importance of recognizing this vulnerability and prioritizing their safety. This means ensuring transparency in the development of AV algorithms and prioritizing the protection of pedestrians and cyclists in unavoidable accident scenarios.

**Advocacy.** From the perspective of pedestrians and cyclists, the correct course of action would involve advocating for ethical and compassionate approaches to autonomous vehicle technology. They believe that protecting everyone's lives, no matter who they are, is the only right way to handle these situations. This aligns with the principles of care ethics, which emphasizes the responsibility to prevent harm and protect the vulnerable (Nair, 2014). Care ethics proposes a shift away from utilitarian calculations towards a more comprehensive approach that considers the vulnerable.

**Implications.** By prioritizing the safety of pedestrians and cyclists, AVs' algorithm can embody care and empathy, potentially leading to a significant reduction in accidents and injuries for these vulnerable road users. According to Rothenbucher et al. (2016), pedestrians and cyclists represented a significant portion of traffic fatalities in 2012, accounting for 16% of all deaths. However, there could also be potential losses, prioritizing care ethics in AV programming might initially deter some buyers and potentially slow down the adoption of this technology, delaying the potential benefits it offers. such as reduced traffic congestion and emissions.

### **Student Position**

While reducing overall harm is still a critical goal, the safety of vulnerable road users, such as pedestrians and cyclists, is equally important. Unlike vehicles' passengers, pedestrians and cyclists lack the protective cocoon of airbags and crumple zones, making them more susceptible to injuries or death in accidents. My position leans more toward the manufacturer's side. AVs should be programmed to make life-or-death decisions. I recognize the immense potential of AV technology to alter transportation and enhance road safety. Therefore, I believe the best approach involves balancing the utilitarian goal of minimizing harm with the ethical responsibility to protect vulnerable road users, as emphasized by the care ethics framework.

Autonomous vehicles are expected to significantly enhance traffic safety by reducing the impact of human error, which accounts for more than ninety percent of crashes (Othman, 2021). Moreover, AVs can improve mobility for people with disabilities, older individuals, and those unable to operate a vehicle. The widespread adoption of AVs promises to revolutionize transportation, eventually making it safer and more efficient for everyone.



To address the ethical dilemma surrounding self-driving cars' decision-making, I recommend a hybrid approach that combines the safety-maximizing goals of utilitarianism with the care and responsibility principles of care ethics. This approach requires programming self-driving cars to prioritize the safety of vulnerable road users in most cases even if it means sacrificing the vehicle or its passengers. However, in cases where multiple lives are at stake, the car could be programmed to minimize overall harm, following a utilitarian approach. Additionally, it is essential for AV algorithms' decision-making to be transparent and open to public discussion, ensuring that these decisions align with the society's values.

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